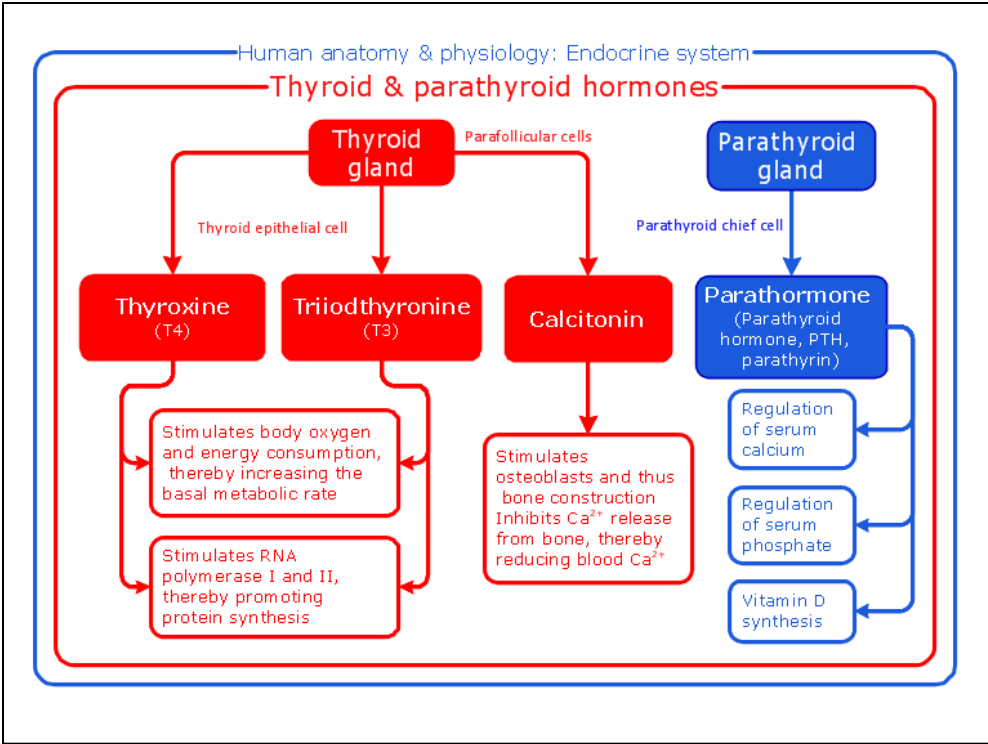


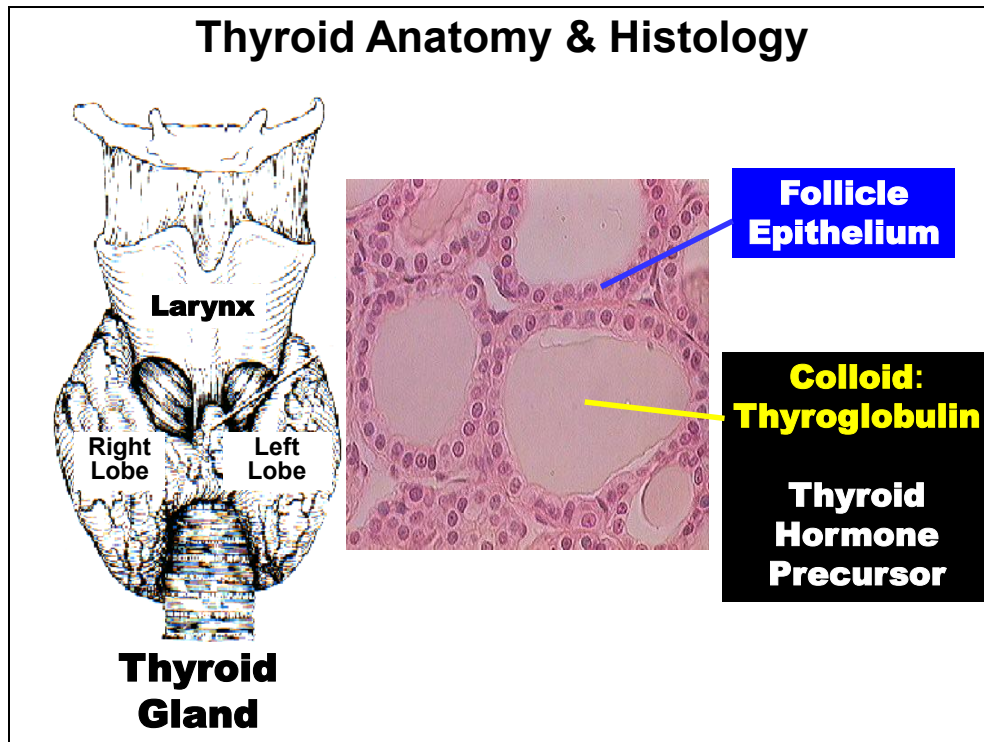
Narrated PowerPoint Presentations

- 1. Overview & Principles of Endocrine System**
- 2. Hypothalamic Neuroendocrine Hormones**
- 3. Anterior Pituitary Hormones**
- 4. Posterior Pituitary Hormones**
- 5. Thyroid Hormones Basic Properties**
- 6. Thyroid Hormone Disorders**
- 7. Parathyroid Hormone**
- 8. Vitamin D and Calcitonin**
- 9. Adrenal Cortex Hormones**
- 10. Steroid Hormone Biosynthesis & CAH Disorders**
- 11. Adrenal Medulla Hormones**



Schematic summary of physiological actions of thyroid and parathyroid hormones.

Thyroid Hormones



The two lobes of the human **Thyroid Gland** (*left*) are located midway between the apex of the thyroid cartilage and the suprasternal notch. A pyramidal lobe is sometimes present.

Histologically, the thyroid gland is made up of multiple spherical acini of varying diameter termed **Thyroid Follicles** (*right*). Each follicle is surrounded by a single layer of **Follicle Epithelium** (*darker ring*) and is filled with a pink-staining proteinaceous material called **Colloid** (*inner area*).

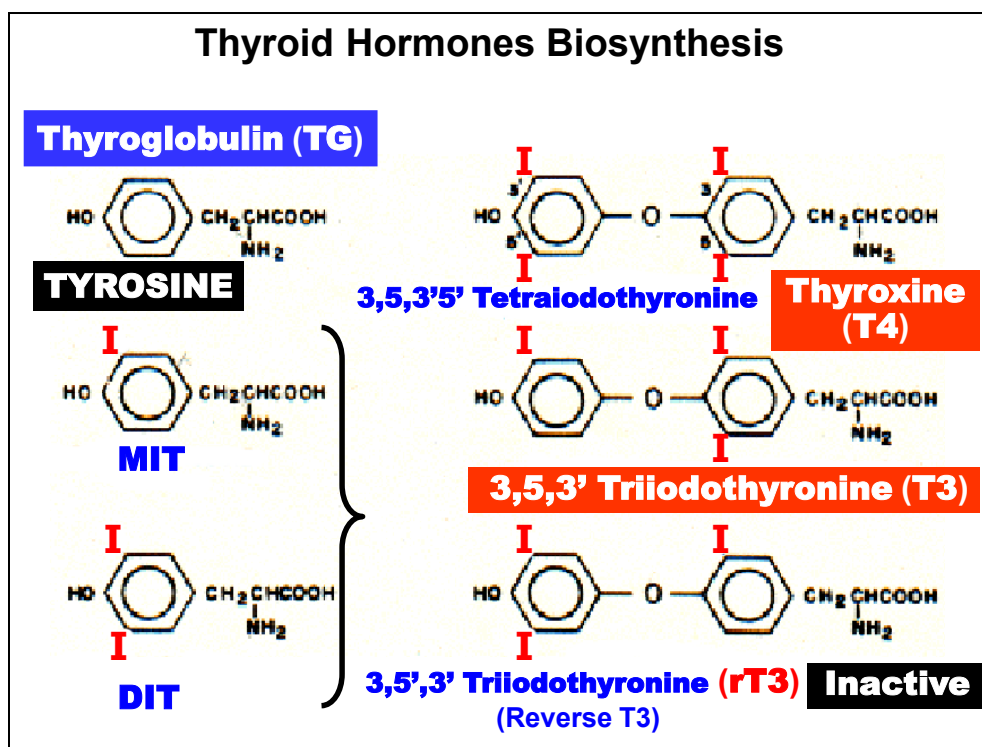
The major colloid component is **Thyroglobulin**, which is a 660 kD dimeric glycoprotein that serves as the **Thyroid Hormone (T4 & T3) Precursor** or substrate.

Thyroid Hormone Properties

- **Classification:** Amino-Acid Derived (Iodinated Tyrosine Dimers)
- **Biosynthesis:** Thyroid Follicles (Thyroglobulin-TPO-T4/T3)
- **Storage:** YES Thyroid Colloid; Plasma Bound Pool
- **Secretion:** Passive Diffusion
- **Transport:** Bound (>99%)
- **Metabolism:** Many Pathways (Long Half-Life: DAYS)
- **Mechanism:** Intracellular Membrane Receptor: Nuclear
- **Actions:** ↑ Metabolic Rate (Calorigenic); Energy Metabolism
- **Regulation:** Hypothalamic-Ant. Pituitary Axis (TRH/TSH/T4)
- **Disorders:** Excess: Graves Disease
Deficiency: Myxedema; Neonatal Hypothyroidism

Thyroid Hormone endocrine properties

Highlights are summarized above.



(top left) Thyroid hormones are Amino Acid-Derived hormones produced from **TYROSINE** residues of the precursor **Thyroglobulin (TG)**, which is secreted into the follicular colloid.

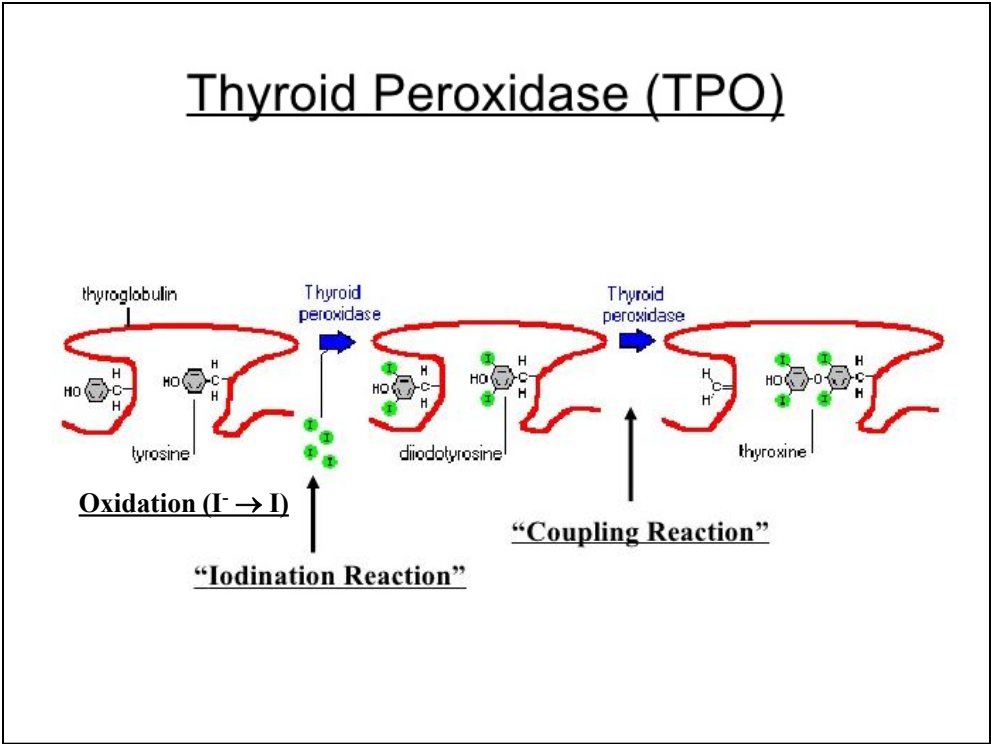
(bottom left) Thyroid hormones are formed from the condensation (or coupling) of two iodinated tyrosine derivatives, Monoiodotyrosine (**MIT**) and/or Diiodotyrosine (**DIT**), which are generated after iodinization of TG (explained in detail later).

(right) Following endocytosis and proteolytic cleavage, within follicular cells, two biologically **active** thyroid hormone derivatives are produced:

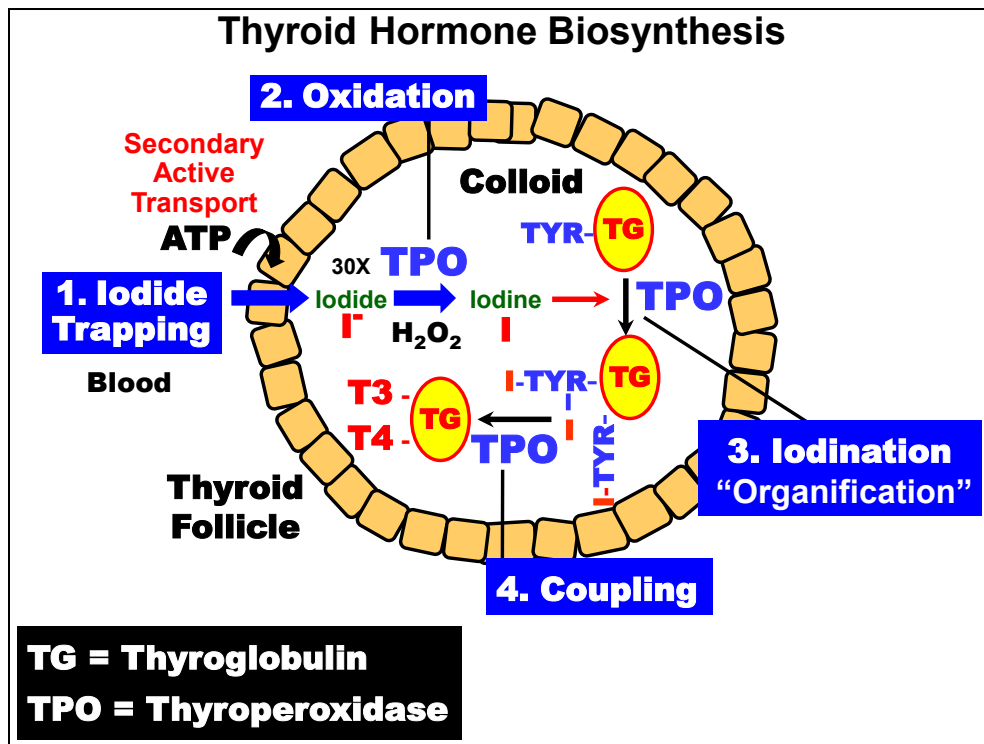
3, 5, 3', 5', Tetraiodothyronine (T4). Also known as **Thyroxine**.

3, 5, 3' Triiodothyronine (T3)

In addition, the product **3, 5, 3' Triiodothyronine**, termed **reverse T3 (rT3)** is also generated, but is functionally **Inactive**. Its levels are sometimes used diagnostically.



Summary of actions of Thyroid Peroxidase in thyroid hormone biosynthesis.



The biosynthesis of **T4** and **T3** by the thyroid gland involves four major steps that require the presence of the thyroid hormone precursor protein, **Thyroglobulin (TG-Tyr)** and the catalytic enzyme **Thyroperoxidase (TPO)**, which acts at several synthesis steps. The absence of TPO results in the complete loss of thyroid hormone production.

NOTE: **Thyroglobulin** is a large glycoprotein (MW 660,000) synthesized within thyroid follicular epithelial cells (*small squares in a circle*) and then secreted by exocytosis into the colloid. TG-Tyr possesses between 100-140 tyrosine residues (not unusually high), which are iodinated and coupled into thyroid hormones. **TPO** is a membrane bound glycoprotein (MW 102,000). **TPO** catalyzes the **oxidation** of I^- to I ; **iodination** of TG, and **coupling** of iodinated tyrosine residues to form thyroid hormones.

(left) **1. Iodide Trapping.** **Secondary Active Transport of Iodide (ATP)** into thyroid follicular cells from blood. This occurs independent of TPO actions.

(top) **2. Oxidation of Iodide to Iodine.** In order to be incorporated into TG-Tyr residues, transported **iodide (I^-)** is rapidly oxidized to **iodine (I)** by hydrogen peroxide (H_2O_2) within the thyroid cell through the catalytic action of **TPO**.

(right) **3. Iodination of TG-Tyr.** The addition of iodine molecules to TG-Tyr is a process termed "**Organification**" and is also catalyzed by **TPO**. Iodination may occur only at the 3 position of a TG-Tyr, forming the intermediate **Monoiodotyrosine (MIT)** or iodinated at the 3 & 5 positions to form **Diiodotyrosine (DIT)**.

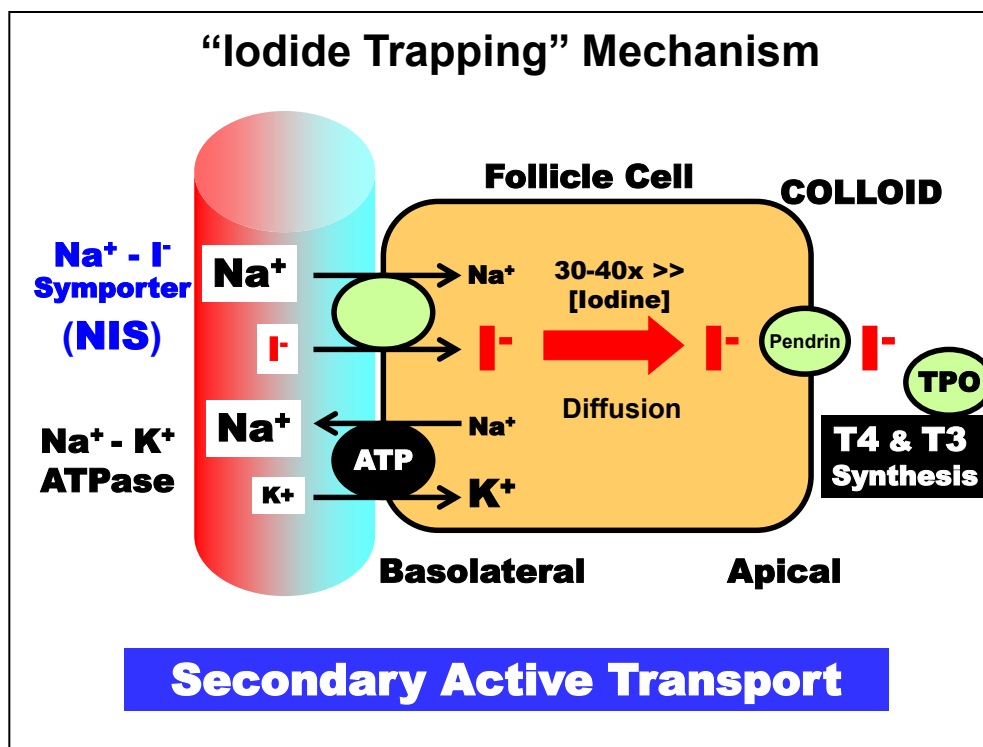
(bottom) **4. Coupling of Iodotyrosyl Residues.** Thyroid hormones (**TG-T3/T4**) are formed through an oxidative **coupling** reaction between two iodinated tyrosine residues (**MIT & DIT**). Coupling results in the elimination of the alanine side-chain from the molecule that forms the outer ring. Coupling of iodotyrosyl residues is catalyzed by **TPO**. Major products formed from the iodination and coupling steps are:

T4 (3, 5, 3', 5', Tetraiodothyronine = Thyroxine) = two molecules of **Diiodotyrosine (DIT)**

T3 (3, 5, 3' Triiodothyronine) = one molecule each of **Monoiodotyrosine (MIT)** and **Diiodotyrosine (DIT)**.

Reverse T3 (rT3) (3, 5', 3' Triiodothyronine) is formed from the coupling of **MIT** and **DIT** but with

MIT on the inside (retaining alanine side chain). **rT3** is **NOT** functionally active, nor are **MIT** & **DIT**.



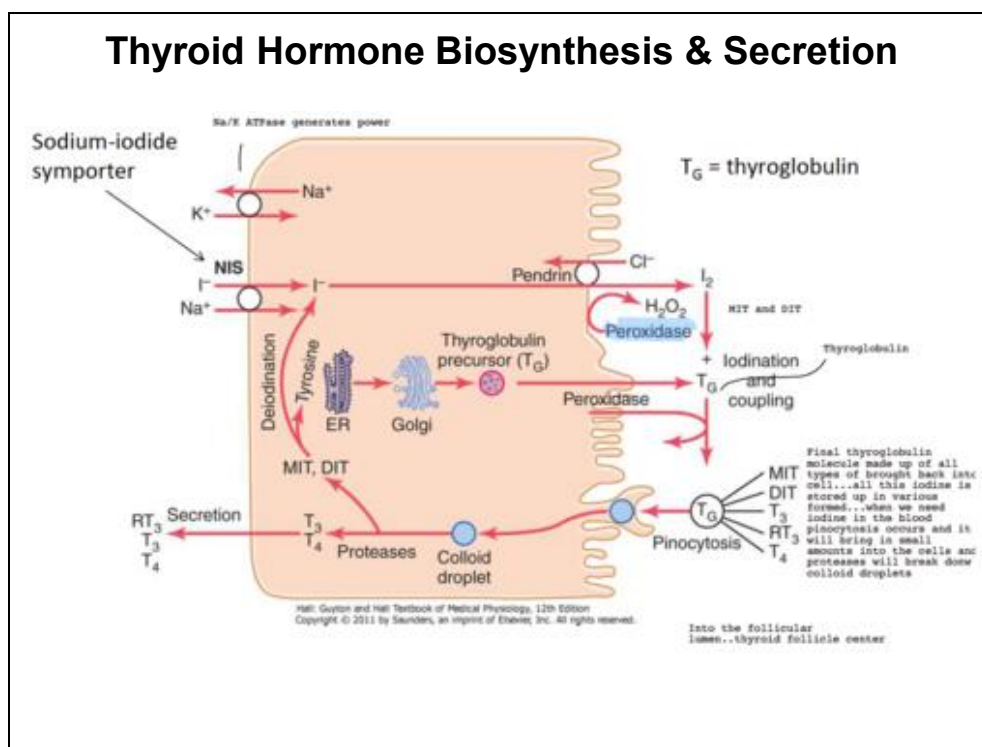
Plasma Iodide (I^-) (*left*) is transported across the Basolateral Membrane of thyroid Follicle Cells (*right*) by a Secondary Active Transport mechanism involving the primary action of a membrane **Na^+/I^- Symporter (NIS)** (*top oval*).

The driving energy is generated by energy-dependent basolateral **$\text{Na}^+ - \text{K}^+$ ATPase** actions, which form an inward directed Na^+ Gradient that drives the transport of plasma I^- into the follicle cell *against* its concentration gradient.

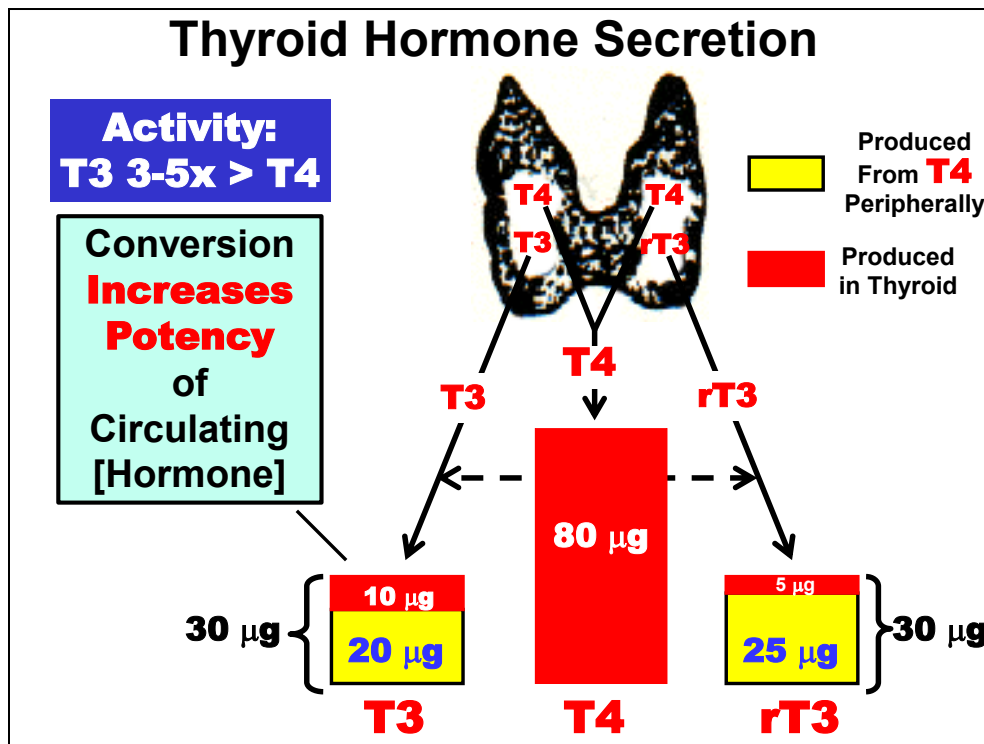
Free intracellular iodide is normally concentrated 30-40x relative to plasma by this mechanism. When the thyroid gland is maximally active, the concentration difference can rise to as high as 250x. The high intracellular I^- forms a Diffusion gradient (*thick leftward arrow*) across the **Apical** membrane into the **Colloid** where thyroid hormone production (T3 & T4 Synthesis) occur under the catalytic actions of an enzyme called Thyroperoxidase (**TPO**).

The iodide channel across the apical membrane is termed the **Pendrin**

transporter, which apparently exchanges I^- for Cl^- across the barrier into colloid.



A more detailed depiction of the dynamics of thyroid hormone biosynthesis. In particular, note the mechanism of thyroid hormone secretion (*bottom*) involving pinocytosis.



The human thyroid gland secretes about 80-100 µg per day of total thyroid hormone (T3 & T4). As indicated by the boxes above, the **majority (≈85%) of daily thyroid hormone secreted from the thyroid gland** (dark area of boxes) is **MOSTLY T4** (middle box-80 µg). Only a small proportion is T3 (left box-10 µg) and the remainder rT3 (right box-5 µg).

However, note that total circulating levels of T3 (30 µg) & rT3 (25 µg) are significantly greater than their secreted amounts.

The difference (light area of boxes) is produced through **deiodination of T4 to T3 & rT3** peripherally in the liver and kidneys by tissue **Deiodinase** enzymes .

(top left text box) The physiological importance of this conversion is that **T3 possesses 3-5x greater functional Activity (or potency) relative to T4.**

Therefore, an elevated rate of T4 to T3 conversion **Increases the Potency** of the overall circulating thyroid hormone concentration in blood.

Thyroid Hormone Transport in Blood

Hormone	[Total Plasma]	% Bound	[Free]
Thyroxine (T4)	8 µg/dL	99.98	<2 ng/dL
Triiodothyronine (T3)	0.15 µg/dL	99.8	<0.3 ng/dL

- **Free T3/T4 = Active Pool of T3/T4**
- **Bound T3/T4 = Unavailable Storage Form (Weeks)**
- **Thyroid Hormone Binding Globulin (45-65% of Bound)**
- **Half-Life is Long: T4 (7 days) > T3 (3 Days)**
- **T4 Prescribed Therapeutically (e.g. 1° Hypothyroidism)**

Most of the circulating active thyroid hormone is **Thyroxine (T4)** as indicated above. Note that both **T3** and **T4** are almost entirely **Bound** (>99%) by plasma proteins because of their hydrophobic polar nature. Less than **1%** circulates in the **Unbound** form.

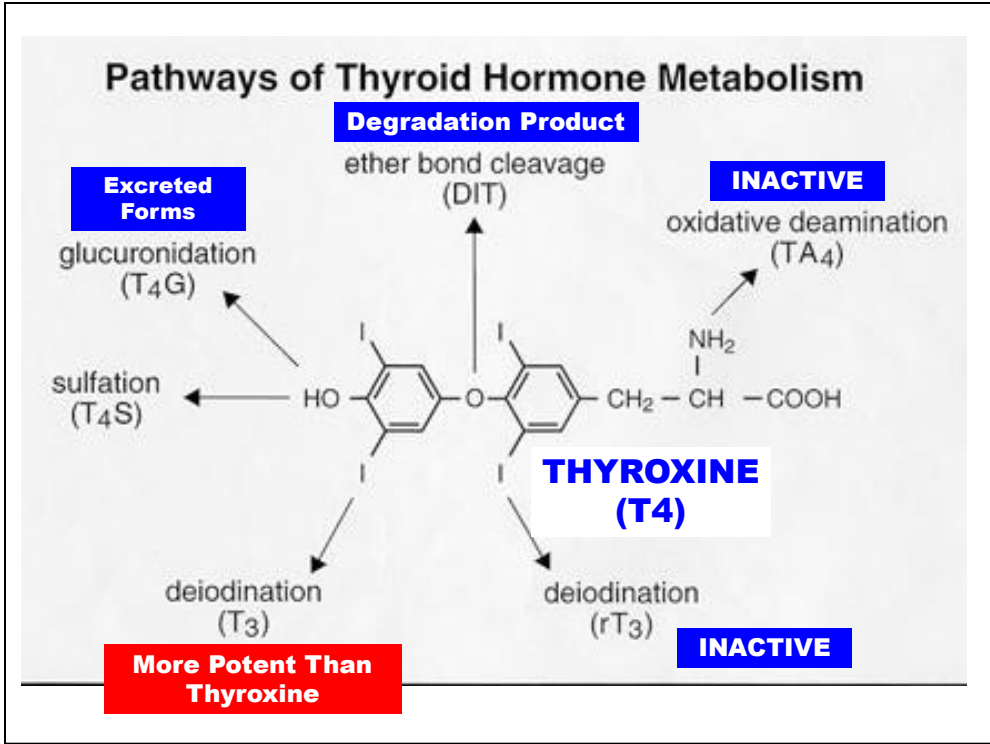
Recall that only the unbound **Free T3 & T4** is physiologically **Active**. However, **Bound T3 & T4** provides an important circulating **Storage Form** of hormone (Several Week Supply) that is independent of glandular secretion and rapidly mobilized.

The major plasma binding protein is the thyroid hormone-specific **Thyroid Hormone Binding Globulin (THBG 45-65% of bound)**.

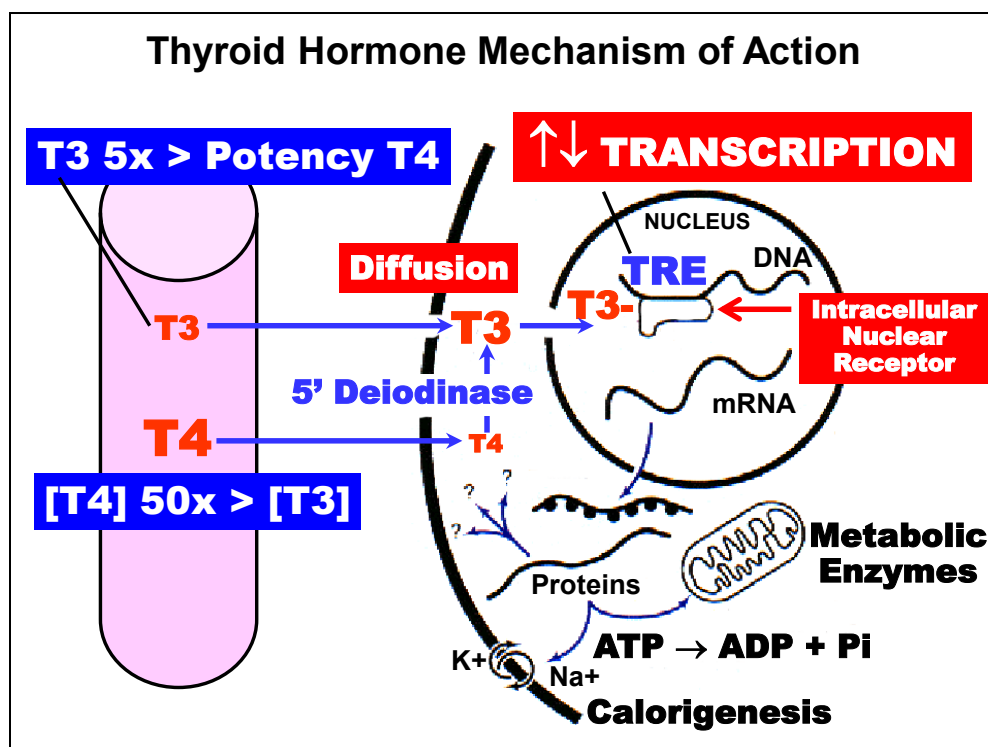
Clinically important is the fact that thyroid hormones have a relatively **Long Half-Life**. Note that **T4 has a longer (>) Half-Life (≈7 days) compared to T3 (≈3 days)**. This is the result of a greater binding affinity of T4 than T3 to THBG.

This is one reason why **T4 is most often prescribed therapeutically for Hypothyroidism** even though T3 is the more physiologically potent hormone. Thus, less exogenous hormone is required and potency is retained

because T4 eventually becomes metabolized to T3.



A reminder of the metabolism of thyroid hormones introduced earlier.



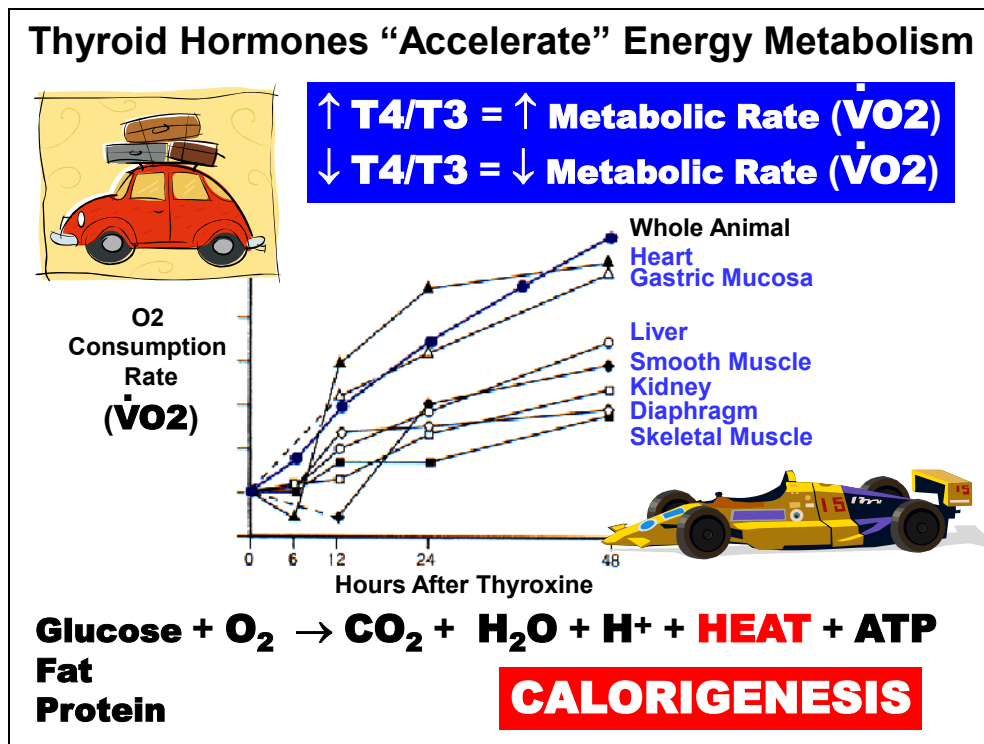
Plasma thyroid hormones (T4 & T3) (left) enter target cells (right) by **Passive Diffusion** (rightward arrows) due to their hydrophobic polar properties. Because of its much higher blood concentration ($[T4] 50x > [T3]$)-bottom left), **most of the total circulating thyroid hormone is in the form of T4.**

However, **almost all entering T4 is converted to T3** (upward arrow) by a **cellular 5' Deiodinase**.

This step largely accounts for the **5x > Potency** of **T3** compared to **T4**. Some have suggested that T4 may basically serve as a prohormone for T3. However, T4 can also apparently bind to receptors, although with lower affinity than T3.

Thyroid hormones produce their physiological actions on peripheral target tissues through **INTRACELLULAR NUCLEAR RECEPTORS**. T3 (or T4) diffuses into the nucleus where it binds as monomers, homodimers or heterodimers to **Nuclear Receptors** (right middle) bound to specific regulatory sequences (TRE) on genes.

Thus, thyroid hormones mainly affect ($\uparrow\downarrow$) the **TRANSCRIPTION** of genes related to its physiological actions [e.g. Metabolic Enzymes; Na⁺-K⁺ ATPase (Calorigenesis)-bottom right].



Although thyroid hormones are NOT absolutely essential for life, their actions are important for the maintenance and optimization of many metabolic and growth processes.

As indicated by the graph above, thyroid hormones (Thyroxine-*x-axis*) increase the **Metabolic Rate** of almost all metabolically active tissues through stimulation of **O₂ Consumption Rate (VO₂-y-axis)**.

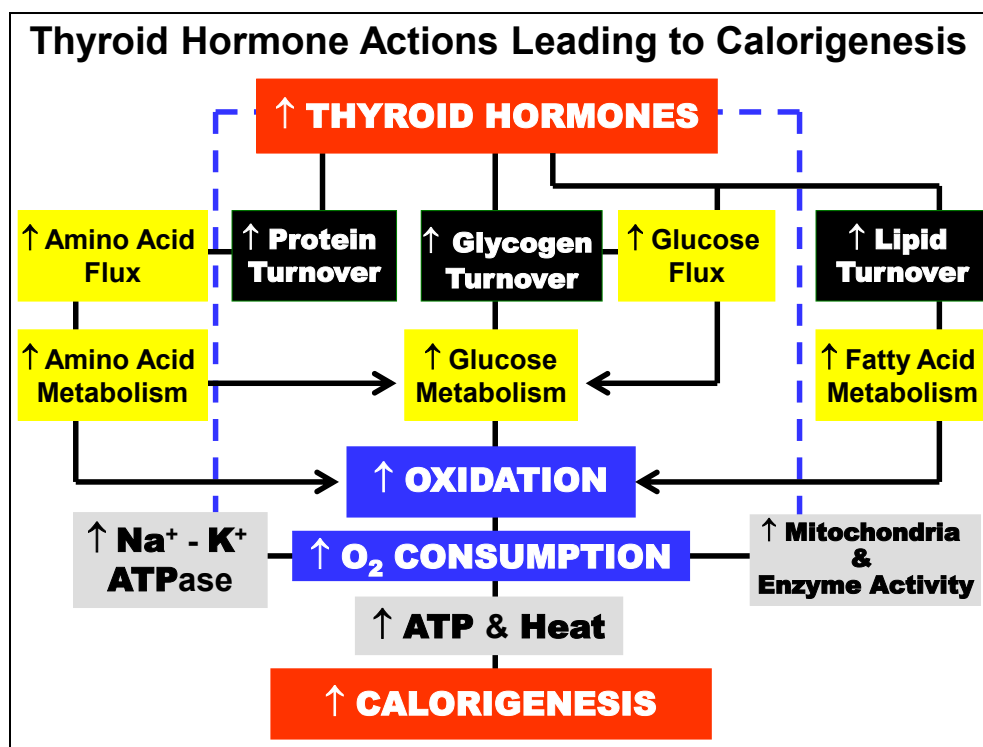
Exceptions include the adult brain, retina, lungs, gonads, uterus, lymph nodes, spleen & anterior pituitary (*not shown*).

Basal Metabolic Rate (BMR) can be increased 60-100% above normal when large quantities of hormone are secreted.

As predicted from the metabolic equation (*bottom*), a fundamental consequence of thyroid hormone action on energy metabolism is **Calorigenesis** (**HEAT** production).

In terms of their effects on metabolic rate, thyroid hormones might be thought of as analogous to the “accelerator” of an automobile, which regulates engine

power by controlling the rate of gasoline consumption for combustion.



Thyroid hormone stimulated **Calorigenesis** (*bottom*) results from its combined actions on many cellular and molecular processes summarized above (*other known mechanisms are not shown*).

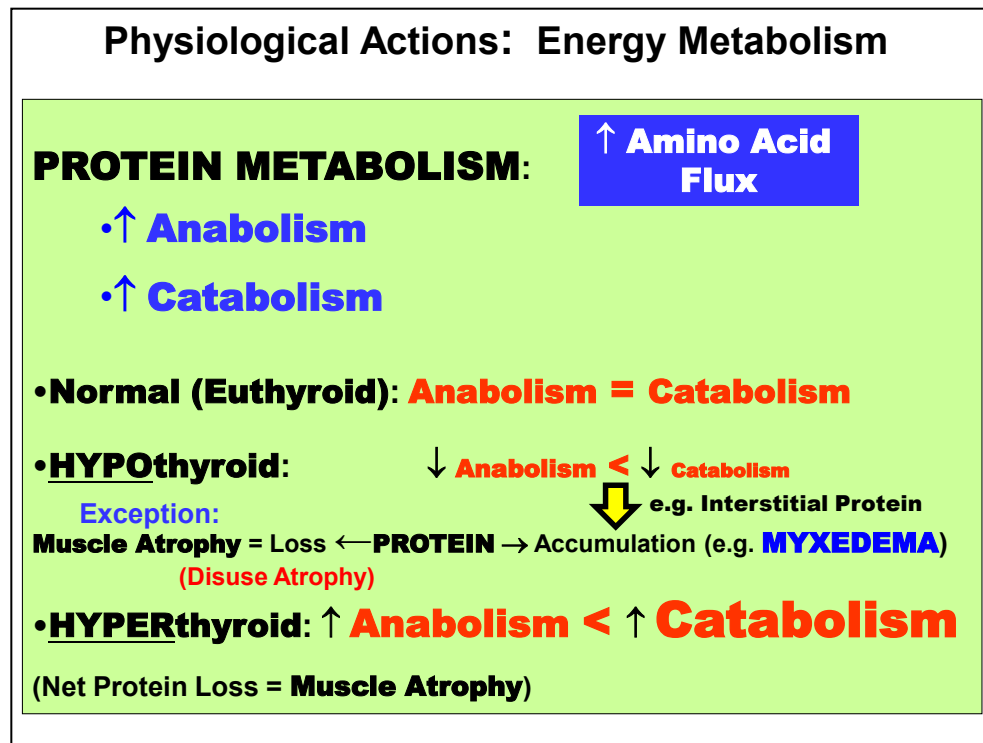
A major action is that thyroid hormones increase the metabolic turnover of the major energy stores (Glycogen, Lipid, Protein).

This results in increased Flux and Metabolism of mobilized energy substrates (Amino Acids, Glucose & Fatty Acids).

Complementary to these metabolic actions, thyroid hormones specifically increase membrane Na⁺-K⁺ ATPase synthesis and activity (*lower left*) in many tissues and promotes Mitochondria production & Enzyme Activity (*lower right*). Thus, increasing the metabolic machinery for increased metabolism

All of actions stimulate flux through Oxidation pathways, which increases O₂ Consumption and results in elevated ATP production. The associated metabolic Heat production accounts for thyroid hormone's increased Calorigenesis effect.

Other actions of thyroid hormone on specific organ systems are summarized in the next two figures. Generally, thyroid hormones have a **STIMULATORY** effect on most processes related to the organ systems.

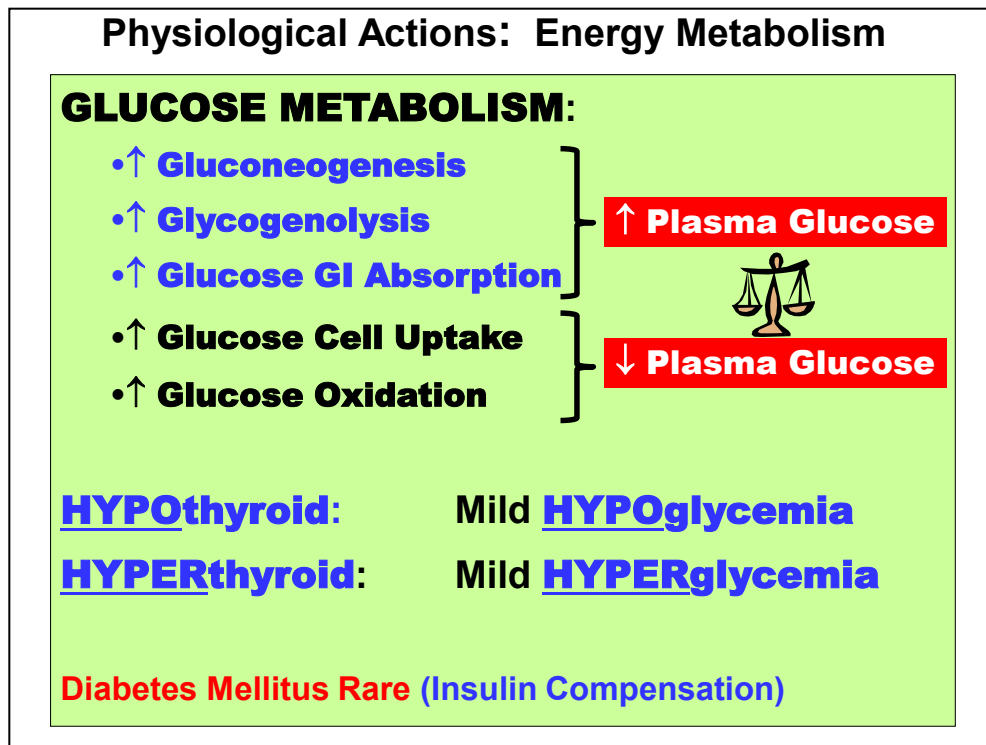


Protein Metabolism. Physiological levels promote **BOTH** protein **Anabolism & Catabolism**, which enhances (↑) Amino Acid Flux without net protein loss or gain.

Hypothyroidism slows overall protein turnover (i.e. ↓ **Anabolism AND** ↓ **Catabolism**) but with greater suppression of Catabolism relative to Anabolism resulting in selective Protein Accumulation (e.g. **Myxedema**). Associated skeletal muscle loss is an exception that results from muscle inactivity (**Disuse Atrophy**) as well as decreased muscle protein anabolism.

Hyperthyroidism stimulates protein turnover, but with greater enhancement of Catabolism, leading to **Net Protein Loss** e.g. Muscle

Atrophy).



Glucose Metabolism. Enhances most processes involved in the production, transport, storage and metabolism of glucose as listed above.

Hypothyroidism decreases net glucose production and mobilization, commonly resulting in Mild **HYPOglycemic** .

Hyperthyroidism increases net glucose production and mobilization, commonly resulting in Mild **HYPERglycemic** effect.

Physiological Actions: Energy Metabolism

LIPID METABOLISM:

↑ **Cardiovascular Disease Risk**

- ↑ **Lipogenesis**; ↑ **Lipolysis**
- ↑ **LDL Uptake**; ↑ **Cholesterol Synthesis**
- **HYPOTHYroid**: ALL ↓, w/ **Net Lipogenesis** = ↓ FFA
 - ↓ Lipolysis = **Weight Gain**
 - ↓ LDL Uptake = ↑ Cholesterol & TGs = **Atherosclerosis**
 - ↑ **Cardiovascular Disease Risk**
- **HYPERthyroid**: ALL ↑, w/ **Net Lipolysis**
 - ↑ LDL Uptake = ↓ Cholesterol & TGs = ↓ CV Risk...but
 - ↑ Lipolysis = ↑ Plasma FFA = **Atherosclerosis**
 - ↑ **Cardiovascular Disease Risk**

Lipid Metabolism. Physiologically, promotes adipose tissue turnover through increased Lipogenesis & Lipolysis, LDL Uptake & Cholesterol Synthesis.

Hypothyroidism SLOWS all lipid metabolic processes, but with **Net Lipogenesis**. Although net lipogenesis decreases serum FFA levels this outcome and the concomitant **decrease in Lipolysis** results in greater fat accumulation and **Weight Gain**, which increases CV disease risk. Moreover, plasma Cholesterol and Triglyceride (TGs) levels increase due to **decreased LDL Uptake**. Overall, this results in and increased risk of **Atherosclerosis**.

Hyperthyroidism STIMULATES all lipid metabolic processes, but with **Net Lipolysis**. Although increased LDL uptake generally results in decreased serum Cholesterol and Triglycerides levels, Plasma FFA levels are elevated due to increased lipolysis increasing the risk for developing **Atherosclerosis**.

Thus, both conditions result in ↑ **Cardiovascular Disease Risk** although through different metabolic effects.

Growth & Development	
Most Organ System Functions Stimulated by T4/T3	
Target Tissue	Physiological Effect(s)
Nervous System	•Fetal/Neonatal Brain Dev. (Congenital Hypothyroidism) Adult Neural Processing
Heart	•+ Chronotropic & + Inotropic; ↑ β -Adrenerg. Receptors ↑ Catecholamine Sensitivity; ↑ Fast Myosin Isoform
Sk. Muscle	•Muscle Contractility; Muscle Mass (Protein Turnover); Contractile Protein Isoforms
Skeletal (Bone)	•Normal Bone & Cartilage Development & Maturation in Concert w/ Growth H.-IGF-1; PTH, Vitamin D
Pathophysiology of Hyper- or Hypothyroid Disorders?	

NERVOUS SYSTEM. Fetal-Neonatal Brain Development. There is a critical period during the perinatal period when thyroid hormone is essential for normal nervous system and brain development. A deficiency during this period results in irreversible brain damage (**Congenital or Neonatal Hypothyroidism**) that is NOT restored with TH replacement. Post-developmental functions of TH include a role in Neural Processing. **Hypothyroidism** leads to mental sluggishness and slowing of muscle reflexes. **Hyperthyroidism** causes tremor, hyperkinesia, restlessness and irritability. Nervousness, anxiety, extreme worry and paranoia are also observed.

CARDIOVASCULAR SYSTEM. Thyroid hormones have a positive (+) **Inotropic & Chronotropic** effect on the heart caused by enhancement of cardiac β -Adrenergic Receptors, which increases **Catecholamines Sensitivity**. These actions enhance major hemodynamic parameters such as blood pressure and volume as well as cardiac output in response to increased metabolism. **Hypothyroidism** decreases heart rate and force of ventricular contractility. **Hyperthyroidism** increases heart rate and contractility.

SKELETAL MUSCLE. Thyroid hormones affect Muscle Contractility (force & velocity) directly and indirectly through their effects on Muscle Mass due to Protein Turnover actions. Thyroid hormones may be anabolic or catabolic on protein metabolism. **Hypothyroidism** causes muscle weakness, cramps and stiffness, resulting in sluggish contractions with slow relaxation times. Manifestations often result in muscle loss due to inactivity (Depressed Motor Neuron Processing), as well as from decreased Protein Synthesis. Interestingly, other some non-muscle proteins accumulate (Myxedema). **Hyperthyroidism** also results in muscle weakness, especially chronically, but in this case muscle loss is mainly due to the relative enhancement of protein Catabolism vs. anabolism.

SKELETAL SYSTEM. Thyroid hormone stimulates increased bone turnover by promoting bone resorption as well as bone formation. Thyroid hormones also act synergistically with other hormones related to skeletal maintenance (GH, IGF-1, PTH, Vitamin D). **Hypothyroidism** retards growth rate in children. Bone growth is slowed and epiphyseal closure delayed. **Hyperthyroidism** abnormally stimulates rapid growth rate. Children become precociously tall, but adult height may be below average due to early epiphyseal closure and bone maturation. Chronically, bone loss and hypercalcemia may occur.

Growth & Development	
Target Tissue	Physiological Effect(s)
GI Tract	•Gastric Secretion; GI Motility; Carbo. Absorption
Respiratory	•Rate & Depth of Breathing (2° to O ₂ Consumption)
Skin	•Cutaneous Vasoconstriction & Vasodilation
Endocrine	•Hormone Production & Metabolism (Half-Life)
Reproductive	•Reproductive Functions; Menstrual Cycle; Erection
Pathophysiology of Hyper- or Hypothyroid Disorders?	

GASTROINTESTINAL SYSTEM. Thyroid hormones influence Gastric Secretion & GI Motility. They also promote GI Glucose Absorption and possibly other metabolites.

Hypothyroidism is associated with loss of appetite and constipation and usually weight gain.

Hyperthyroidism is associated with increased appetite and more frequent defecation and typically diarrhea along with weight loss. Weight changes in either condition are variable due to parallel effects on metabolic rate.

RESPIRATORY SYSTEM. Thyroid hormone affects the respiratory system secondary to its metabolic effects. Increased or decreased metabolic rate affect the rate of O₂ Consumption, which modifies the Rate & Depth of Ventilation. **Hypothyroidism** suppresses ventilation (↓ rate & depth) secondary to ↓ metabolic rate. **Hyperthyroidism** stimulates ventilation (↑ rate & depth) secondary to ↑ metabolic rate.

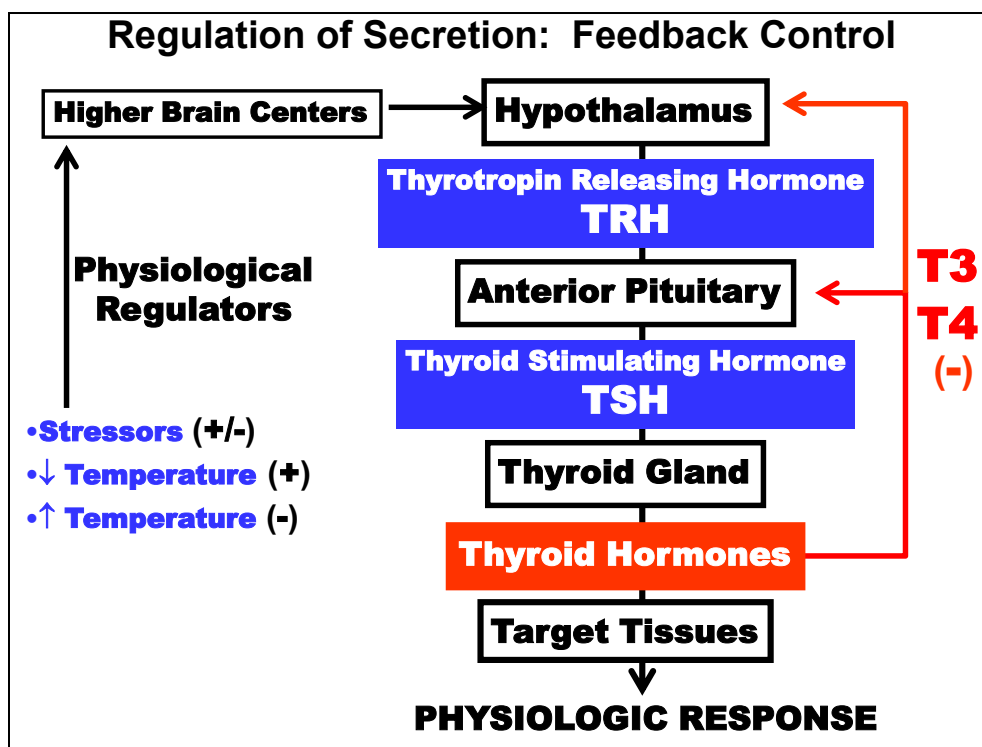
SKIN. Thyroid hormone causes cutaneous Vasoconstriction or Vasodilation.

Hypothyroidism enhances cutaneous Vasoconstriction. Skin is cool and pale and sensitive to cold. **Hyperthyroidism** enhances cutaneous Vasodilation. Skin is warm and moist due to excessive sweating and heat intolerance.

ENDOCRINE SYSTEM. Thyroid hormone affects Hormone Production (Secretion) & Metabolism (Half-Life) of many hormones. This affects hormone secretion, plasma levels and target tissue actions since many are regulated by negative feedback mechanisms.

Hypothyroidism causes decreased endocrine hormone metabolism. **Hyperthyroidism** causes increased endocrine hormone metabolism. Endocrine hormone activity is variable due to integrated and feedback effects.

REPRODUCTIVE SYSTEM. Thyroid hormone is required for normal Reproductive Functions. **Hypothyroidism** causes decreased fertility and irregular menstruation in women. In men, loss of libido. **Hyperthyroidism** causes long menstrual cycles and sometimes the loss of menstruation (amenorrhea) in women. In men, impotence.



Synthesis and secretion of thyroid hormone depends on the stimulation of thyroid follicular cells by **Thyroid Stimulating Hormone (TSH)**, which is secreted by the Anterior Pituitary Gland. In the absence of TSH, thyroid cells are quiescent and atrophy.

Secretion of TSH, in turn, requires stimulation by **Thyrotropin Releasing Hormone (TRH)** which is secreted from the Hypothalamus. TSH and thyroid hormone production become impaired in its absence.

(left) Physiological Regulators of TRH production and secretion include certain types of **Stressors** and **Body Temperature**.

Negative Feedback Control (right) Constant plasma levels of T3 & T4 are achieved by a classical Feedback Control mechanism. T3 & T4 mainly act through inhibition (-) of TSH secretion from the Anterior Pituitary (*solid arrow*) but may also act at the level of the Hypothalamus (*dashed arrow*)

High T4 & T3 serum levels (Hyperthyroidism) inhibit TSH synthesis & release.

Low T4 & T3 serum levels (Hypothyroidism) stimulate TSH synthesis & release.

A patient is diagnosed with an endocrine disorder classified as a **Primary Hyperthyroid** condition caused by an adenoma of the thyroid gland.

Which set of laboratory results below is most likely associated with this condition?

- A. Low T4; Low TSH**
- B. High T4; Low TSH**
- C. High T4; High TSH**
- D. Low T4; High TSH**

A patient is evaluated for a possible thyroid gland disorder. Laboratory results reveal the following data:

- Total T4 = 1.5 $\mu\text{g/dL}$ (Normal = 5.5-11.0 $\mu\text{g/dL}$)
- Free T4 = 0.2 ng/dL (Normal = 0.8-2.7 ng/dL)
- TSH = 12.4 $\mu\text{U/ml}$ (Normal = 0.4-4.2 $\mu\text{U/ml}$)

What is the appropriate classification of this condition?

A. Primary Hyperthyroidism

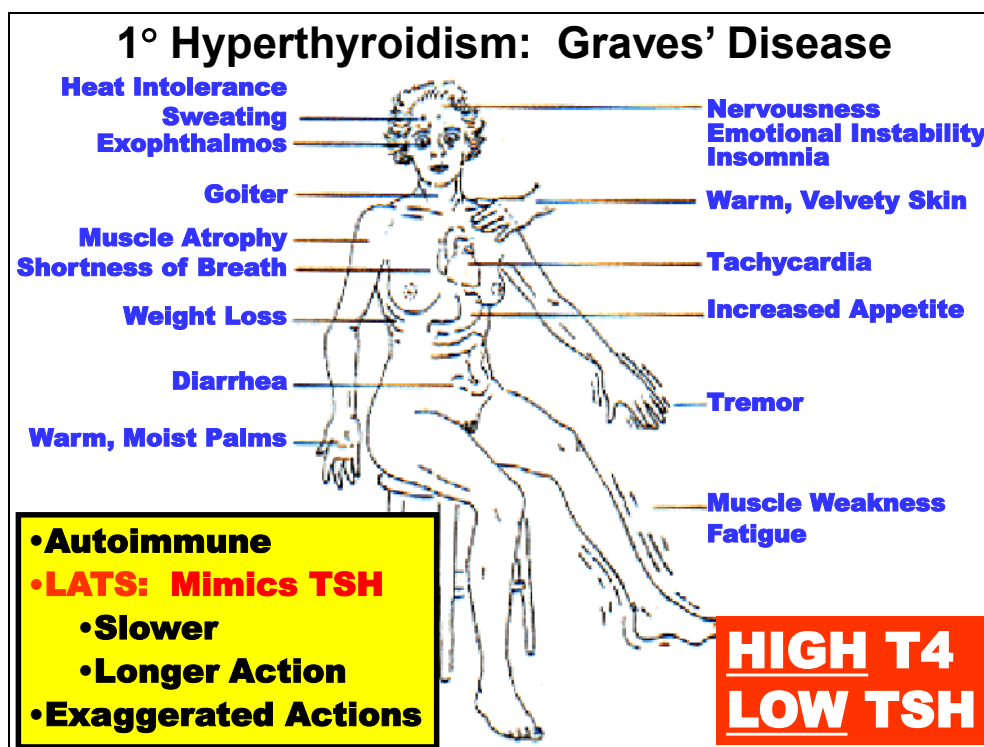
B. Primary Hypothyroidism

C. Secondary Hyperthyroidism

D. Secondary Hypothyroidism

Narrated PowerPoint Presentations

- 1. Overview & Principles of Endocrine System**
- 2. Hypothalamic Neuroendocrine Hormones**
- 3. Anterior Pituitary Hormones**
- 4. Posterior Pituitary Hormones**
- 5. Thyroid Hormones Basic Properties**
- 6. Thyroid Hormone Disorders**
- 7. Parathyroid Hormone**
- 8. Vitamin D and Calcitonin**
- 9. Adrenal Cortex Hormones**
- 10. Steroid Hormone Biosynthesis & CAH Disorders**
- 11. Adrenal Medulla Hormones**



(bottom left text box) **Graves' Disease** is an Autoimmune Disease causing diffuse thyroid enlargement (diffuse toxic goiter) & hormone hypersecretion. The abnormality results from **Hyperstimulation** of thyroid follicle cells and is caused by an antibody against thyroid antigens called Long-Acting Thyroid Stimulator (LATS) (or TSI = Thyroid Stimulating Immunoglobulin). There are probably several LATS-like IgG's. LATS acts by mimicking TSH, except that its action is Slower and of Longer duration.

The clinical manifestations of Graves' disease all result from the **Exaggerated Actions** of the normal physiological functions of thyroid hormones. Common manifestations in young adults are summarized above. Cardiovascular & Myopathic manifestations are more prominent in the Elderly. Children uniquely display accelerated growth w/ premature bone maturation (short stature).

(bottom right text box) Diagnostically, abnormally elevated levels of T3 & T4 (**HIGH T4**) circulating in plasma inhibit TSH secretion, resulting in **LOW TSH** levels that are a typical laboratory findings related to Graves' Disease.

Pathophysiology of Graves' Disease:

(Exaggerated Thyroid Actions)

CALORIGENESIS: NEURON EXCITABILITY:

- Sweating
- Warm Skin
- Moist Palms
- Heat Intolerance
- Nervousness
- Hyperactivity & Fatigue
- Muscle Tremor
- Diarrhea

INCREASED BMR:

- Weight Loss
- Increased Appetite
- Tachycardia
- Shortness of Breath

LATS AUTOMIMUNE:

- Exophthalmos
- Goiter

PROTEIN CATABOLISM:

- Muscle Atrophy
- Muscle Weakness & Fatigue

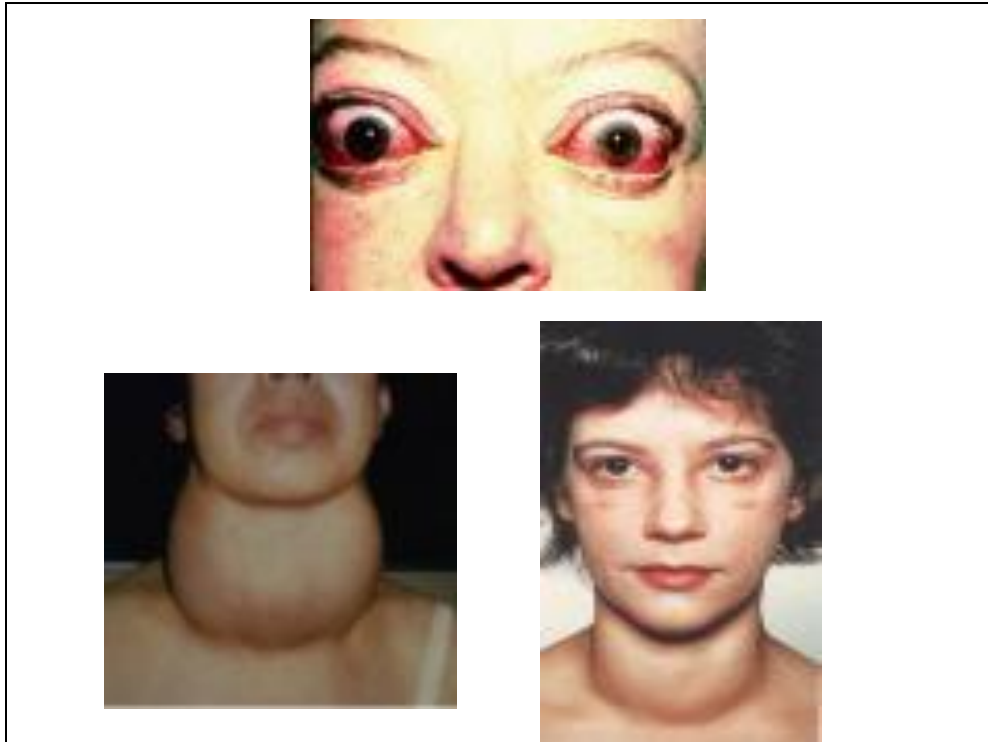


Clinical manifestations of Graves' disease are categorized above in relation the major physiological functions of thyroid hormones (see Physiological Functions section). Some symptoms are idiopathic.

(*bottom right*) Graves' disease is characteristically accompanied by **Exophthalmos** (exaggerated eye protrusion).

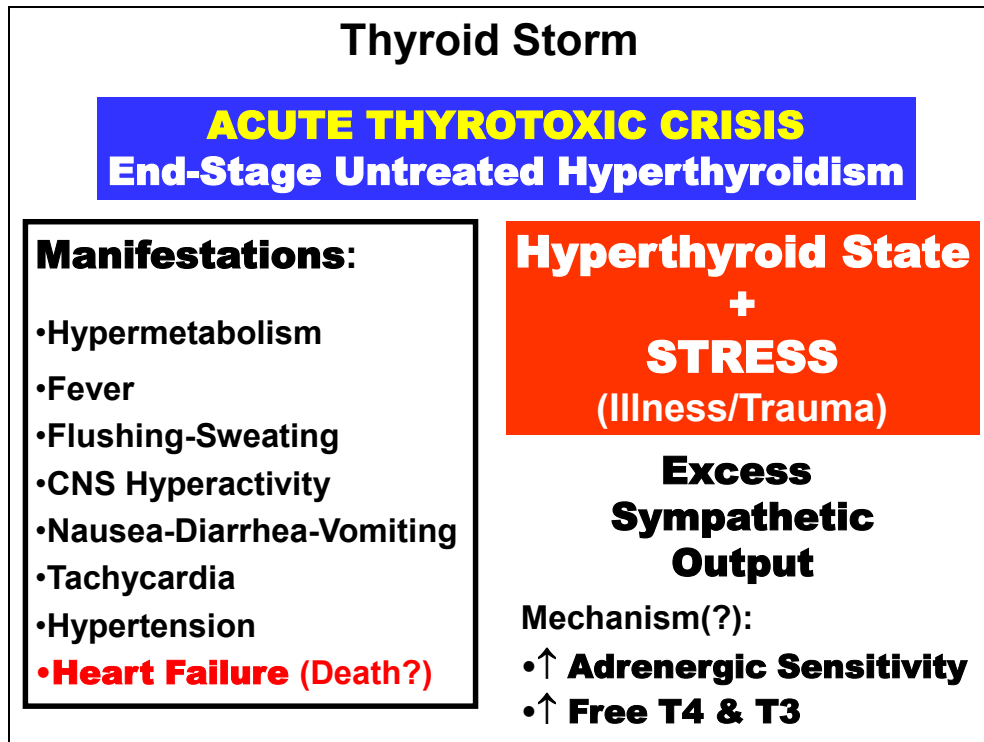
Exophthalmos is due to swelling of extraocular muscles and connective tissue within the rigid bony walls of the orbit. This pushes the eyeball forward. The cause of the exophthalmos is almost certainly an autoimmune attack on the extraocular muscles and orbital connective tissues. Antibodies are apparently formed to antigens common to the eye muscles and the thyroid. The condition responds to glucocorticoids or other immunosuppressive agents.

Goiter is the consequence of the excessive unregulated production of thyroid hormone, whose colloidal synthesis and storage rate outpaces the rates of thyroglobulin proteolysis and secretion from the gland. Note that goiter development is not specific to Graves' disease or hyperthyroid conditions in general as this feature also presents with hypothyroid disorders.



Images of **Exophthalmos** (top) and **Goiter** (bottom) associated with Graves Disease.

Note that goiter also develops with other thyroid conditions including hypothyroid disorders although due to different etiologies.



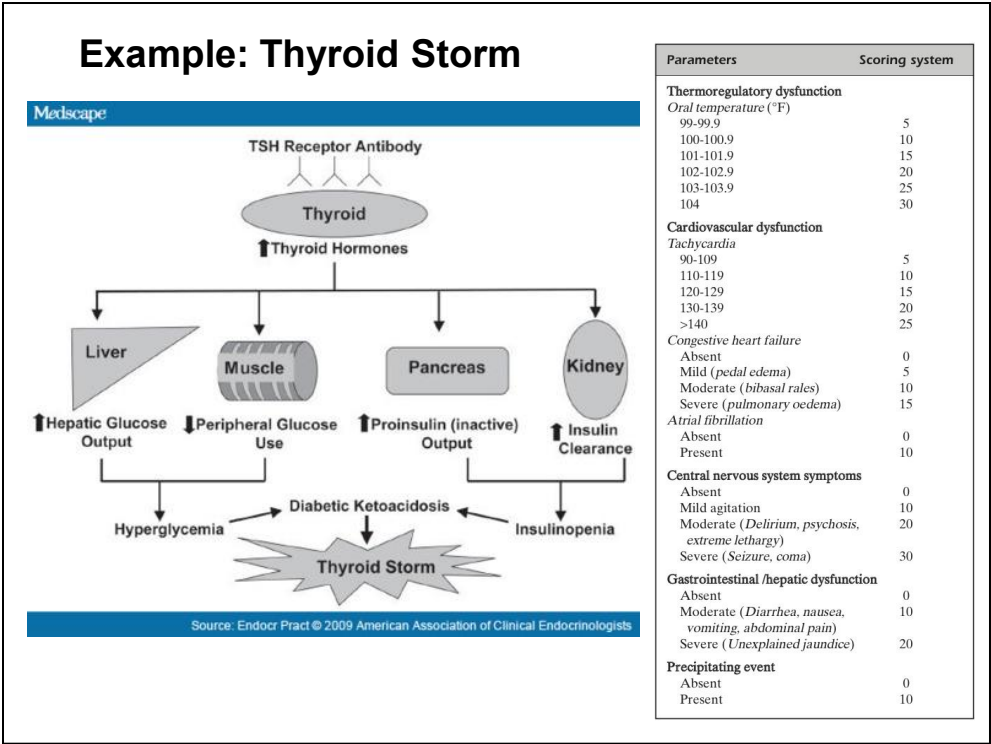
Thyroid disease is a debilitating condition, but is not usually lethal.

However, chronic hyperthyroidism (e.g. Graves' Disease) can sometimes acutely and severely decompensate into a condition termed **Acute Thyrotoxic Crisis** or more commonly **Thyroid Storm**.

The precipitation of this condition apparently occurs when a chronic hyperthyroid state is coupled (+) with an acute **Stress** situation such as Illness (e.g. infection) or Trauma (e.g. surgery).

It is thought that the Excess Sympathetic Output associated with stress greatly exaggerates the already hyperthyroid state of the patient. Mechanistically, it has been proposed that the combination of increased tissue Adrenergic Receptor Sensitivity and an unusually elevated amount of Free T4 & T3 may mediate the acute manifestations.

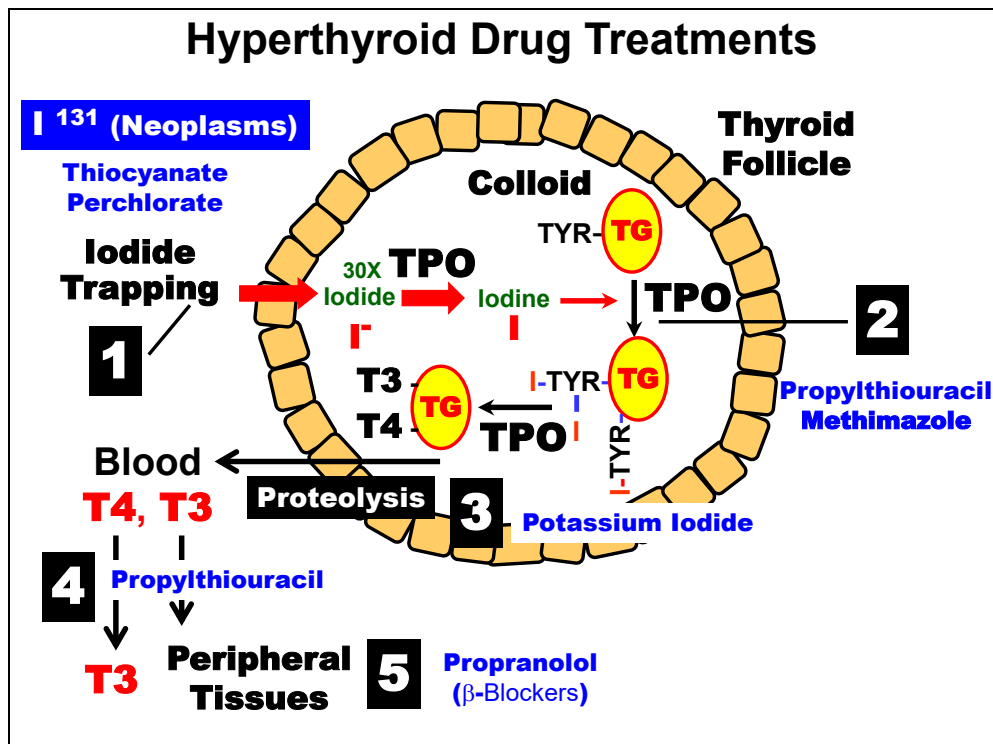
These are summarized above (*lower left*). This highly exacerbated hyperthyroid condition may sometimes result in Death, usually due to Heart Failure if left untreated.



FYI only.

Example of pathophysiology leading to **Thyroid Storm** (left).

Clinical approach to diagnosing patient with potential **Thyroid Storm** (right).

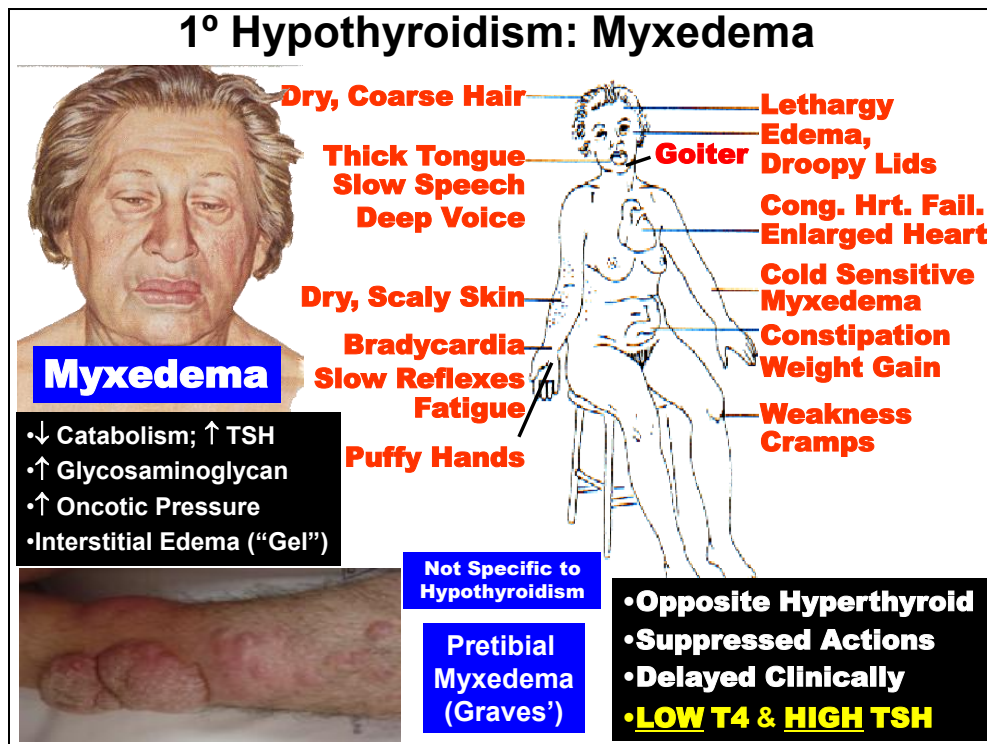


In addition to Graves' disease, hyperthyroidism can also be caused by other conditions including: Excess secretion of TSH due to hypophyseal tumors (Secondary Hyperthyroidism); Toxic nodular goiter or adenomas of the thyroid (Primary Hyperthyroidism); overzealous treatment of hypothyroid patients with therapeutic thyroid hormone.

Pharmacologically, hyperthyroidal conditions can be treated by a variety of drugs that act to suppress thyroid hormone secretion at several steps related to their biosynthesis or metabolism. These include inhibiting iodide transport (**1. Iodide Trapping**), preventing organification of thyroglobulin via TPO inhibition (**2**), suppressing cleavage of TG and secretion of T4 & T3 (**3. Proteolysis**), preventing T4 conversion to active T3 (**4**), and suppressing TH actions on peripheral tissues (**5**).

Specific classes of drugs will be presented in more detail next year during the Pharmacology course. In addition to pharmacological interventions, Surgical Ablation of the thyroid gland may be a necessary option (e.g. Graves' Disease).

Iodine I^{131} is also used for to destroy hyperfunctioning or neoplastic thyroid cells



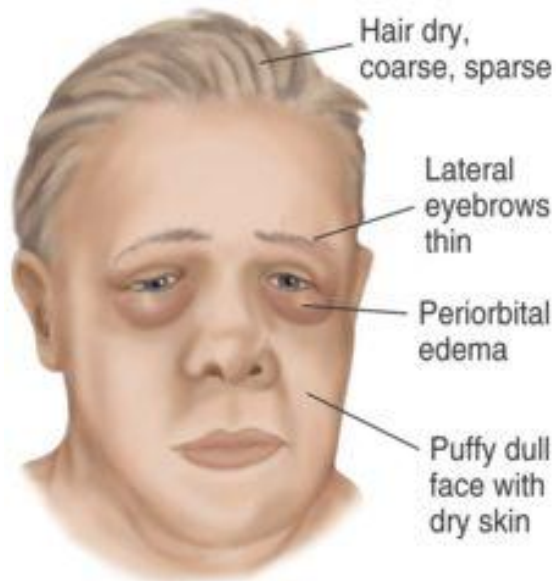
Hypothyroidism results from **deficient** thyroid hormone secretion (e.g. iodine deficiency = endemic goiter) or target cell action. (*right text box*) Clinical manifestations are basically **Opposite Hyperthyroidism**, causing generalized **Suppressed Actions** of thyroid hormones. Hypothyroidism in infants and children also results in slowing of growth & development, including mental retardation.

The classical **Primary Hypothyroid** condition in adults is **Myxedema** (*above*), which is a characteristic clinical manifestation resulting from the suppressed (↓) Protein Catabolism effect of hypothyroidism. **Excess Glycosaminoglycan** deposition in extracellular spaces (mainly skin & muscle) creates (↑) Oncotic Pressure leading fluid accumulation and the formation of a thick "Gel-like" **Interstitial Edema** that gives the patient an outstanding puffy and rough facial appearance termed Myxedema (*top left*).

Another classical manifestation of hypothyroidism is the development of an enlarged thyroid (**Goiter**) caused by the chronic and excessive stimulation of the thyroid gland by TSH. TSH secretion is high because **Negative Feedback Inhibition** by thyroid hormones to the anterior pituitary is reduced. The increase in size is largely due to increased thyroglobulin synthesis, which remains uniodinated in the colloid (e.g. iodine deficiency).

In addition to myxedema & goiter, other common features are summarized above. Try to correlate these symptoms to the known physiological actions of thyroid outlined earlier.

Classic Features of Myxedema



Classic clinical features of Myxedema

Graves': Pretibial Myxedema

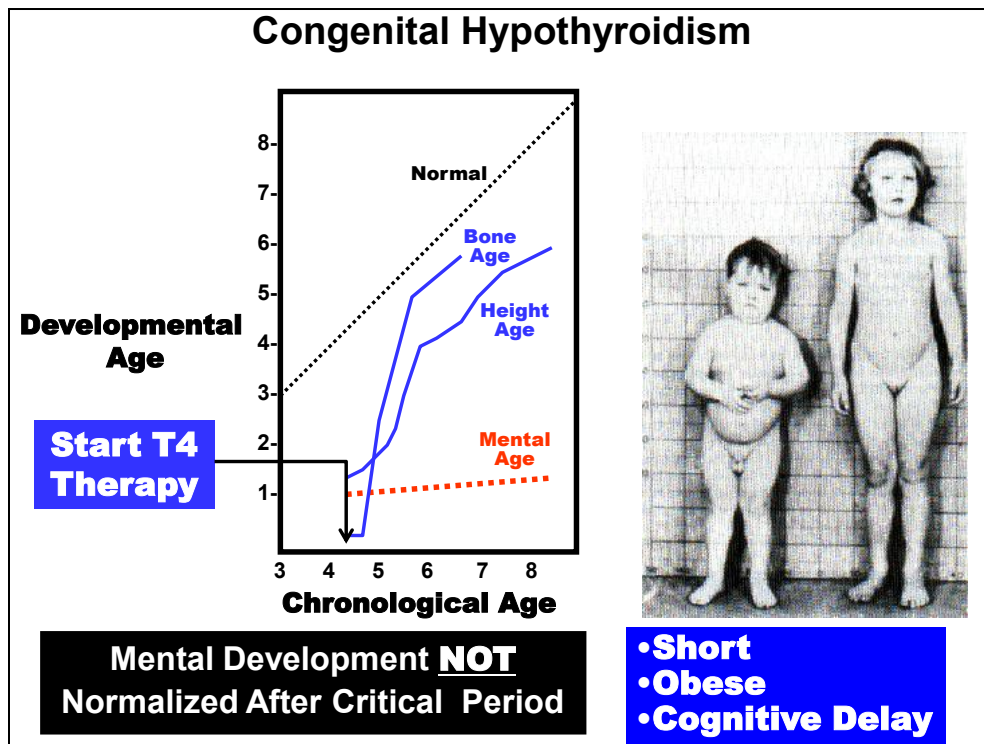


**Pretibial Myxedema
(Graves' Disease)**



Myxedema development is not restricted to hypothyroid conditions.

A relatively rare (1-5%) development with Graves' disease is a condition termed **Pretibial Edema** (Graves' Dermopathy) which results from increased deposition of glycosaminoglycans stimulated by TSI.



Children who are **Hypothyroid** from birth were historically termed “Cretins” (Cretinism) which is now considered archaic and the term longer used.. Appropriate terms is **Congenital Hypothyroidism**.

Clinical manifestations in newborns include respiratory difficulty, cyanosis, jaundice, poor feeding, hoarse cry, umbilical hernia, and marked retardation of bone maturation. As shown above (*right*), children who are hypothyroid from birth (*male on left*) are typically Dwarfed (**Short Stature**) compared to normal adolescents (*female on right*) with characteristic Potbellies (**Obese**). **Cognitive Delay** is a diagnostic trait along with laboratory findings. A serum T4 under 6 mg/dL or a serum TSH over 30 mg/dL is indicative of **Congenital Hypothyroidism**. Diagnosis confirmed by radiological bone age.

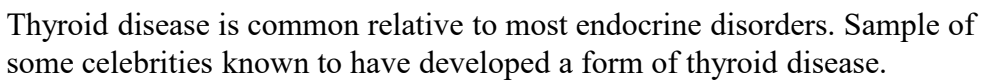
As discussed earlier, mental retardation always results if thyroid hormone therapy is not instituted within a critical period during neonatal or early infant brain development. As shown above (*top left*), children diagnosed with hypothyroidism after the “critical period” respond to thyroid hormone intervention (Start T4 Therapy) in most physical traits (Bone Age, Height Age), but not in mental capacity (Mental Age).

Etiologies. Congenital abnormalities of the hypothalamic-pituitary axis that cause goiter can cause congenital hypothyroidism . Alternatively, neonatal hypothyroidism may result from failure of the thyroid to descend from its origin at the base of the tongue during embryonic development. In newborns and children, deficiency generally results from lack of iodine in endemic areas (endemic hypothyroidism).

Congenital Hypothyroidism



Images of **Congenital Hypothyroid** condition in young child (left) and adult (right)



Narrated PowerPoint Presentations

- 1. Overview & Principles of Endocrine System**
- 2. Hypothalamic Neuroendocrine Hormones**
- 3. Anterior Pituitary Hormones**
- 4. Posterior Pituitary Hormones**
- 5. Thyroid Hormones Basic Properties**
- 6. Thyroid Hormone Disorders**
- 7. Parathyroid Hormone**
- 8. Vitamin D and Calcitonin**
- 9. Adrenal Cortex Hormones**
- 10. Steroid Hormone Biosynthesis & CAH Disorders**
- 11. Adrenal Medulla Hormones**